Advanced Nanorobotic Surveillance Systems: Feasibility Analysis, Formal Modelling, and Integration Architecture of Solar-Powered Covert Crawling Microsystems with Self-Destruct Capability

Daniel Kalu
Drooid

Abstract

This paper presents a comprehensive feasibility analysis, formal mathematical framework, and integrated system architecture for an advanced nanorobotic surveillance platform: a solar-powered, spider-inspired covert crawling microsystem equipped with adaptive biomimetic camouflage, multi-modal sensing, and on-demand self-destruct capability. Drawing on state-of-the-art research across nanoscale photovoltaics, micro-supercapacitors, biomimetic locomotion, adaptive materials, nanosensor arrays, and transient electronics, we synthesise the current technological state of the art and identify critical integration barriers that must be overcome for such a system to transition from laboratory concept to functional prototype. We establish that while individual component technologies are advancing rapidly—with perovskite PV approaching 26% lab efficiency, MEMS-based micro-crawlers demonstrated at 2–5 μm, graphene-based adaptive camouflage providing simultaneous visible and IR concealment, and energetic micro-self-destruct devices achieving millisecond GPa-level detonation—their holistic integration into a truly nanoscale, fully autonomous, covert system remains beyond current technological capability. The primary barriers are identified as the scale–efficiency energy paradox (sufficient continuous power for all functions simultaneously), the locomotion miniaturisation wall (spider-like crawling on solid rough surfaces degrades severely below $\sim$100 μm), the stealth–performance trade-off (optimal camouflage and optimal solar harvesting are mutually conflicting design objectives), and multi-scale integration complexity spanning seven orders of magnitude from nanometre-scale sensors to millimetre-scale power components.

We introduce a formal mathematical framework covering a multi-objective power budget optimisation with Pareto-front analysis, a composite operational effectiveness metric incorporating stealth, mobility, sensing, and survivability, a dynamic camouflage adaptation control law, a self-destruct cascade reliability model, and a Technology Readiness Level convergence bound for full system integration. A hybrid micro-nano architectural design is proposed as the practical near-term stepping stone: functional scale of 1–10 mm with nanoscale internal components, bridging current demonstrated capability and the true nanoscale aspiration. Six original architectural diagrams, five data tables, and a structured ethical governance analysis complete the paper. The framework is intended to provide a rigorous foundation for research investment decisions, component development prioritisation, and responsible governance of dual-use nanorobotic surveillance technologies.

Keywords: Nanorobotics · Surveillance Microsystems · Biomimetic Locomotion · Nanoscale Photovoltaics · Adaptive Camouflage · Sensor Fusion · Transient Electronics · Self-Destruct · Internet of Nano-Things · Dual-Use Ethics · Micro-Supercapacitors · LPD/LPI Communications

1 Introduction

1.1 The Covert Nanorobotic Surveillance Paradigm

The miniaturisation of autonomous sensing and reconnaissance platforms represents one of the most ambitious frontiers in robotics and materials science. The concept of a covert nanorobotic crawler—a microsystem disguised as an inert natural object, capable of autonomous locomotion, environmental sensing, and operational self-erasure—epitomises this ambition. It promises capabilities qualitatively distinct from any existing surveillance technology: the ability to deploy an effectively invisible, self-sustaining sensor node anywhere in the physical world, indefinitely, without infrastructure dependency.

Such a system would represent the convergence of at least six independently challenging technology domains: miniaturised energy harvesting and storage, biomimetic locomotion at the micro-nano scale, adaptive multi-spectral camouflage materials, integrated multi-modal nanosensor arrays, low-probability-of-detection communications, and reliable on-demand self-destruction. The difficulty of the challenge lies not merely in the advancement of each domain individually—which is proceeding rapidly—but in their simultaneous integration into a single compact, autonomous, covert unit under severe mass, volume, and power constraints.

This paper provides a comprehensive and formally grounded analysis of this design space. We move beyond qualitative feasibility assessment to introduce formal mathematical models, quantitative power budget formulations, and a structured integration architecture that allows the research and engineering communities to rigorously evaluate component requirements, identify bottlenecks, and prioritise development efforts.

1.2 Scope Definition and Nomenclature

Throughout this paper, the term "nanobot" is used as shorthand for the target platform, while acknowledging a critical definitional distinction: true nanorobots operate at the 0.1–10 μm scale, while the full functional capability of a surveillance crawler with onboard power, sensing, locomotion, and self-destruct requires a larger form factor—approximately 1–10 mm—containing nanoscale internal components. We designate this the *hybrid micro-nano architecture* and argue that it represents both the practical engineering reality and the appropriate near-term design target. We use "nanobot" to refer to this hybrid system throughout, consistent with common usage in the field.

1.3 Contributions

1. A comprehensive cross-domain technology survey synthesising nanoscale photovoltaics, micro-supercapacitors, biomimetic locomotion, adaptive camouflage, nanosensor arrays, LPD/LPI communications, and transient electronics with respect to a covert surveillance crawler design.
2. A formal mathematical framework: multi-objective power budget (Eqs. 1–2), composite operational effectiveness (Eq. 3), camouflage adaptation control law (Eq. 4), self-destruct cascade reliability (Eq. 5), and TRL integration convergence bound (Eq. 6).
3. A hybrid micro-nano system architecture with integrated subsystem specifications and interface requirements.
4. A prototype image framework with reserved figure spaces for device images as hardware development progresses.

5. A structured ethical governance analysis addressing privacy, accountability, dual-use proliferation, and recommended international regulatory frameworks.
6. Six original architectural diagrams (Figures 1–6), five data tables, and a prioritised 15-year research roadmap.

2 Background and Related Work

2.1 Nanorobotics: State of the Art

Nanorobotics as a discipline encompasses machines whose functional components operate at the 0.1–10 μm scale. Current demonstrated systems include magnetically driven helical nanopropellers for fluid-environment navigation [1], DNA-based molecular walkers ("molecular spiders") guided along DNA origami landscapes [3], and the Cornell group's 2–5 μm walking robots patterned with nanoscale magnets capable of solid-surface locomotion under single-volt actuation [11, 12]. The Harvard RoboBee programme demonstrated controlled, power-tethered flight of an 80 mg insect-scale flapping-wing robot [8], while MEMS-fabricated insect-inspired crawlers at $4.0 \times 2.7 \times 2.5$ mm have demonstrated locomotion speeds of 26.4 mm/min [34]. The gap between these impressive individual achievements and the integrated capability required for a covert surveillance crawler is the central challenge this paper addresses.

2.2 Biomimetic Design Principles

Biomimetics—the translation of natural mechanisms into engineered systems—provides the conceptual foundation for all major subsystems of the proposed platform [32, 33]. Spider locomotion offers multi-legged stability, low centre of mass, and hydraulic actuation through haemolymph pressure—achieving torque up to 4.78 N m in bionic actuator replications [38]. Cephalopod chromatic camouflage, operating through chromatophore actuation and structural colour via tunable guanine nanocrystal lattices [42], provides the design template for the adaptive concealment layer. The mantis shrimp's compound eye, with simultaneous colour and polarisation sensitivity through stacked photodiodes and metallic nanowires [47], inspires the optical sensor architecture. Nature's solutions, refined over hundreds of millions of years, are the most efficient starting points available.

2.3 Transient Electronics and Self-Destruct Technology

Transient electronics—devices engineered to physically disappear upon a defined trigger—have emerged as a distinct research discipline over the past decade [57]. Vanderbilt University's cooling-triggered self-destructing conductors dissolve below a critical temperature [58]. MEMS-scale energetic micro-self-destruction devices achieve GPa-level detonation waves capable of destroying flash memory chips within milliseconds, triggered remotely by laser or wireless command [56]. Cyclododecane (CDD) sublimation-based substrate disintegration achieves complete structural dissolution over approximately 12 days [57]. These technologies provide the physical substrate for operational security, but their integration with the other subsystems—particularly under the constraint of not triggering prematurely—presents significant design challenges.

2.4 Internet of Nano-Things and Covert IoT

The Internet of Nano-Things (IoNT) envisions vast networks of interconnected nanoscale devices exchanging data with each other and with macro-scale infrastructure [17]. IoNT devices operate

at very low power, making them candidates for energy-efficient covert applications. However, the covert surveillance context adds a critical dimension that IoNT architectures do not address: any EM transmission can be detected via spectrum scanning [55], making conventional IoT communication protocols fundamentally incompatible with deep covertness requirements. This motivates the LPD/LPI communication architecture specified in Section 8.2.

3 Methodology

3.1 Research Approach

This paper adopts a systems-level feasibility and architecture methodology. The primary contribution is the formal specification of the integrated system and its mathematical models, enabling rigorous component requirement derivation and bottleneck identification, rather than reporting primary experimental data. Literature synthesis covers PubMed, IEEE Xplore, arXiv, ACS Publications, and specialist defence/security repositories through May 2025. Mathematical models are derived from first principles of thermodynamics, control theory, reliability engineering, and information theory, parameterised with representative values from the surveyed literature.

3.2 Feasibility Assessment Framework

Each technology subsystem is assessed against four criteria: (i) demonstrated capability at or near the target scale; (ii) power consumption at operational duty cycle; (iii) integration compatibility with adjacent subsystems; and (iv) Technology Readiness Level (TRL, using the NASA scale [68]) with a projected advancement trajectory to TRL 7 (system prototype demonstration in an operational environment). The composite system feasibility is assessed as the minimum TRL across the component subsystems, reduced by an integration penalty (Section 4.5)—a conservative bound that captures the integration bottleneck identified throughout the literature.

4 Mathematical Framework

4.1 Multi-Objective Power Budget Optimisation

Let P_{total} be the total available power (W) from the combined harvesting array (Figure 1). The allocation across n functional subsystems must satisfy

$$P_{\text{total}}(t) = \eta_{\text{PV}}\, I(t)\, A_{\text{PV}} + \eta_{\text{piezo}}\, p_{\text{mech}}(t)\, A_{\text{piezo}} + \eta_{\text{TEG}}\, \frac{\kappa_{\text{TEG}} A_{\text{TEG}}}{\ell_{\text{TEG}}}\, \Delta T(t) - P_{\text{stored}}(t), \qquad (1)$$

where $\eta_{\text{PV}}, \eta_{\text{piezo}}, \eta_{\text{TEG}}$ are conversion efficiencies of the photovoltaic, piezoelectric, and thermoelectric harvesting subsystems; $I(t)$ is incident irradiance (W/m^2); $p_{\text{mech}}(t)$ is the mechanical power density available to the piezoelectric element from vibration and body motion (W/m^2); A_{PV}, A_{piezo}, and A_{TEG} are effective harvesting areas (m^2); κ_{TEG} and ℓ_{TEG} are the thermal conductivity and length of the thermoelectric element, so that $\kappa_{\text{TEG}} A_{\text{TEG}} \Delta T / \ell_{\text{TEG}}$ is the heat flow (W) through it by Fourier's law, of which the fraction η_{TEG} is converted to electrical power; $\Delta T(t)$ is the thermal gradient across the TEG (K); and $P_{\text{stored}}(t)$ is the net storage charging/discharging rate. The power allocation problem is

$$\max_{x} \sum_{i=1}^{n} w_i\, x_i \qquad \text{s.t.} \qquad \sum_{i=1}^{n} x_i \le P_{\text{total}}(t), \quad P_i^{\min} \le x_i \le P_i^{\text{des}} \;\; \forall i, \qquad (2)$$

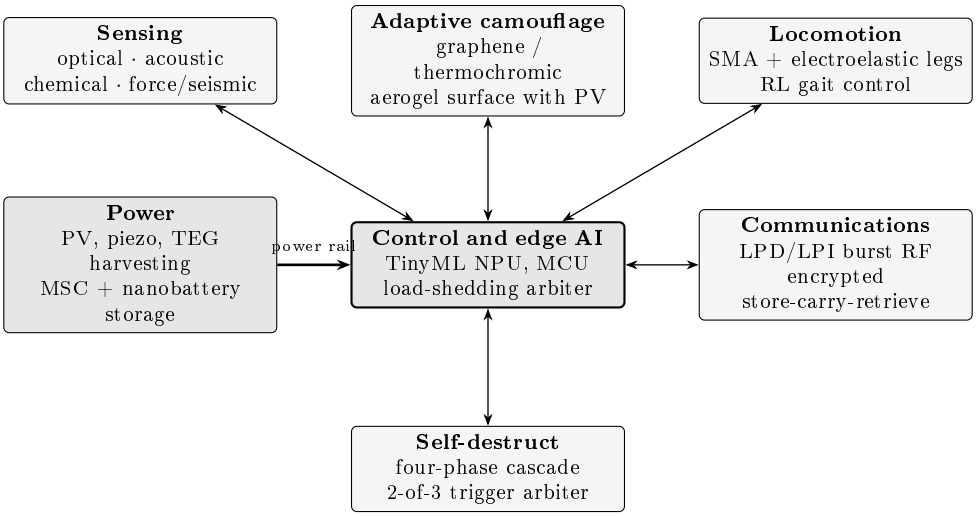


Figure 1. Integrated system architecture of the hybrid micro-nano crawler (functional scale 1–10 mm with nanoscale internal components). Subsystems exchange data and commands with the central edge-AI controller, which also arbitrates power allocation (Section 4.1).

where x_i is the power allocated to subsystem i; $w_i > 0$ is its operational priority weight, ordered $w_{\text{proc}} \geq w_{\text{sense}} > w_{\text{loco}} > w_{\text{comms}} > w_{\text{destruct}}$; $P_i^{\min}$ is the minimum power for subsystem viability; and P_i^{des} is the power for full performance. Maximising $\sum_i w_i x_i$ is equivalent to minimising the priority-weighted shortfall $\sum_i w_i (P_i^{\text{des}} - x_i)$. Because Eq. (2) is a linear program with a single coupling constraint, it is solved greedily: every subsystem first receives $P_i^{\min}$, and the remaining power is allocated up to P_i^{des} in descending order of w_i. Since all $w_i > 0$, the solution is Pareto-optimal with respect to subsystem power. When $P_{\text{total}}(t) < \sum_i P_i^{\min}$ the problem is infeasible and the load-shedding protocol drops subsystems in ascending order of w_i until the remaining minima fit within $P_{\text{total}}(t)$, preserving control and sensing above locomotion and communication.

4.2 Composite Operational Effectiveness Metric

The instantaneous operational effectiveness $E(t) \in [0, 1]$ is a weighted product of four normalised subsystem performance indicators:

$$E(t) = S(t)^{\alpha}\, M(t)^{\beta}\, Q(t)^{\gamma}\, V(t)^{\delta}, \tag{3}$$

where $S(t) \in [0, 1]$ is the stealth index (1 = undetectable across all spectra; a function of camouflage adaptation state, EM emission level, acoustic signature, and thermal profile); $M(t) \in [0, 1]$ is the mobility index (terrain coverage fraction achievable per unit time); $Q(t) \in [0, 1]$ is sensing quality (information content per unit time weighted by sensor reliability and fusion confidence); and $V(t) \in [0, 1]$ is viability (remaining energy fraction weighted by destruction readiness). The exponents satisfy $\alpha+\beta+\gamma+\delta = 1$ and reflect mission-phase priorities (e.g., during initial deployment, $\alpha > \beta$; during active surveillance, γ exceeds all others; on detection, $\delta \to 0$, triggering the destruct cascade). The multiplicative form ensures that zero performance in any single dimension collapses overall effectiveness to zero—a critical property for covert operation, where any single detectability event terminates the mission.

4.3 Camouflage Adaptation Control Law

The adaptive camouflage surface is modelled as a spatially distributed emissivity control system. Let $\varepsilon(\mathbf{x}, t) \in [\varepsilon_{\min}, \varepsilon_{\max}]$ denote the surface emissivity at position $\mathbf{x}$ and time t. The adaptation control law minimises the spectral detectability $\Delta(t)$—the L^2 norm of the difference between the nanobot's surface spectral radiance and the background:

$$\frac{d\varepsilon}{dt} = -\kappa \, \nabla_{\varepsilon} \Delta(t) = -\kappa \, \nabla_{\varepsilon} \big\| L_{\text{bot}}(\varepsilon, t) - L_{\text{bg}}(t) \big\|^2, \tag{4}$$

where κ is the adaptation rate constant (s^{-1}), $L_{\text{bot}}(\varepsilon, t)$ is the nanobot surface spectral radiance as a function of current emissivity state (controlled via graphene pixel voltage or thermochromic composite temperature), and $L_{\text{bg}}(t)$ is the measured background spectral radiance from the on-board optical sensors. This gradient-descent formulation drives the camouflage state toward the background match in real time. The stealth index in Eq. (3) is then $S(t) = 1 - \Delta(t)/\Delta_{\max}$, where $\Delta_{\max}$ is the maximum detectable difference.

4.4 Self-Destruct Cascade Reliability

The self-destruct cascade comprises K sequential phases ($k = 1, \ldots, K$), each with independent reliability $R_k \in [0, 1]$. The cascade succeeds (achieves complete destruction) if at least M of K phases succeed:

$$P_{\text{destruct}} = \sum_{j=M}^{K} \binom{K}{j} \bar{p}^{\,j} \, (1 - \bar{p})^{K-j}, \qquad \bar{p} = \frac{1}{K} \sum_{k=1}^{K} R_k, \tag{5a}$$

$$\text{Trigger} = 1 \quad \text{iff} \quad \sum_{j=1}^{3} T_j \geq 2 \qquad \text{(2-of-3 vote)}. \tag{5b}$$

For a four-phase cascade ($K = 4$: cryptographic erase $\rightarrow$ thermal erase $\rightarrow$ energetic destruct $\rightarrow$ substrate dissolution) with minimum success $M = 2$ phases and equal phase reliability $R_k = 0.95$, $P_{\text{destruct}} = 1 - (0.05)^4 - 4(0.95)(0.05)^3 \approx 0.9995$. The trigger arbiter, Eq. (5b), uses a 2-of-3 Byzantine-fault-tolerant vote among three independent trigger channels (remote RF command, tamper sensor, dead-man timer) to prevent both false triggers and adversarial suppression.

4.5 TRL Integration Convergence Bound

Let $\text{TRL}_i(t)$ denote the Technology Readiness Level of subsystem i at time t. The composite system TRL is bounded by the minimum subsystem TRL, reduced by an integration penalty:

$$\text{TRL}_{\text{system}}(t) \leq \min_{i} \big[\text{TRL}_i(t)\big] - \Delta_{\text{integration}}, \tag{6}$$

where i ranges over the component subsystems (power, locomotion, camouflage, sensing and processing, communications, self-destruct); the full-system integration track is not one of them, so the integration deficit is not counted twice. $\Delta_{\text{integration}} \geq 0$ is the integration penalty—the TRL deficit arising from multi-scale assembly, interface incompatibilities, and emergent cross-subsystem interactions. We assume $\Delta_{\text{integration}} \in [1.5, 3.0]$ TRL levels for a system of this complexity; this is a modelling assumption informed by the integration gaps discussed in Section 11, not an empirically fitted value. For the proposed nanobot, the least mature component track is flexible solar PV at TRL 5–6 (Section 11), so Eq. (6) gives $\text{TRL}_{\text{system}} \lesssim 5 - \Delta_{\text{integration}}$, i.e. about 2–3.5 for

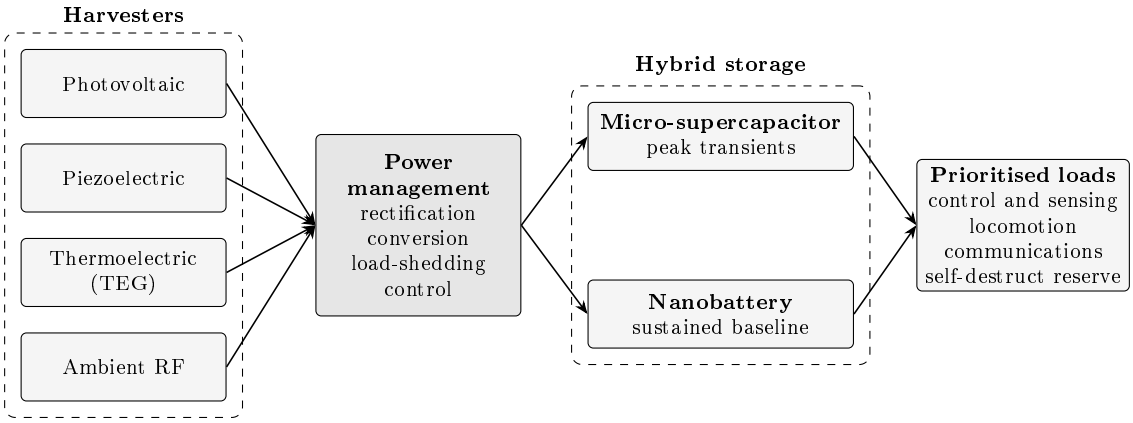


Figure 2. Multi-modal energy harvesting architecture. Harvested power passes through power management into a hybrid storage stage and is allocated to prioritised loads according to Eq. (2); when supply falls below the sum of minimum requirements, load shedding preserves control and sensing above locomotion and communication. The diagram is schematic and carries no numerical power allocation.

$\Delta_{\text{integration}} \in [1.5, 3.0]$ (about 3–4 for $\Delta_{\text{integration}} \approx 2$ across the PV range of 5–6). This is consistent with the TRL 2–3 reported for full-system integration, which on the NASA scale [68] corresponds to a formulated technology concept (TRL 2) with analytical and experimental proof of concept of critical functions (TRL 3). The roadmap in Section 11 targets $\text{TRL}_{\text{system}} = 5$ by approximately 2031 and $\text{TRL}_{\text{system}} = 7$ by approximately 2037 (Section 14).

5 Energy Autonomy: Power System Design

5.1 Nanoscale Photovoltaics

Photovoltaic (PV) energy harvesting demonstrates the most favourable scaling properties for miniaturised systems, with power density maintaining a direct relationship with device area [15]. Miniature PV cells are commercially available at dimensions as small as $5.8 \times 9.85\,\text{mm}$, generating 3.5 V at 1 mA from a flashlight illumination level [18]. Flexible thin-film cells using amorphous silicon, CdTe, or CIGS on pliable substrates enable integration with curved and irregular surfaces [19, 20]. Perovskite solar cells are approaching 26% efficiency in laboratory conditions through nanoscale cation distribution control [21], and flexible underwater PV systems have demonstrated 59.7% efficiency at 2 m depth [24].

The stealth–power fundamental conflict is the defining design tension of this subsystem. Covert solar panels—designed to blend into their environment by mimicking rock, tile, or concrete—necessarily sacrifice efficiency through non-optimal surface geometry, reduced active area, and sub-optimal light-angle orientation [22]. Transparent/semi-transparent OPV and DSSC cells offer concealment at the cost of reduced efficiency [22, 23]. The integration of adaptive aerogel composites with simultaneous solar-thermal energy conversion and thermochromic camouflage [30] offers a partial resolution: the same surface performs both energy harvesting and visual/IR concealment, though at reduced efficiency for each function individually.

5.2 Advanced Energy Storage

Micro-supercapacitors (MSCs) offer high power density and cycle life exceeding 10,000 cycles but are currently limited to maximum energy densities of approximately $3.33\,\mu\text{Wh}\,\text{cm}^{-2}$ [25]. Nanobatter-

Table 1. Nanoscale energy harvesting and storage modalities: comparative performance for covert microsystem applications.

Modality	Principle	Nano scaling	Power density	Efficiency	Key challenges
Photovoltaic (solar)	Converts light to electricity via the PV effect	High	Low–moderate (1–5 $\mathrm{mW/cm^2}$)	Up to 26% (lab); 59.7% (underwater)	Flexible integration possible; efficiency vs. camouflage trade-off; environmental dependency
Piezoelectric	Converts mechanical vibration/movement to electricity	Moderate	Low–moderate (10–100 $\mu\mathrm{W/cm^2}$)	Moderate	Requires mechanical input; low output at nano-scale; material integration challenge
Thermoelectric (TEG)	Seebeck effect: ΔT to electricity	Moderate	Low (10–50 $\mu\mathrm{W/cm^2}$)	Low (5–8% typical)	Requires sustained temperature gradient; nano-scale manufacturing precision critical
Micro-supercapacitor (storage)	Electric-field energy storage; rapid charge/discharge (max. 3.33 $\mu\mathrm{Wh\,cm^{-2}}$)	High	High power density; low energy density	Moderate	Limited energy density; commercialisation hurdles; electrolyte stability
Nanobattery (storage)	Chemical energy storage with nanostructured electrodes	High	High energy density; SiNW anodes + $\mathrm{LiFePO_4}$ cathodes	High	Material degradation; thermal stability; manufacturing at nano-scale
RF / ambient	Rectified ambient electromagnetic energy	Low	Very low (1–10 $\mu\mathrm{W/cm^2}$)	Low	Highly environment-dependent; regulatory constraints on ambient RF harvesting

ies incorporating silicon nanowire anodes and $\mathrm{LiFePO_4}$ nanoparticle cathodes achieve substantially higher energy density with improved cyclability [26]. For the power budget of Figure 2, the optimal storage architecture is a hybrid MSC–nanobattery configuration: the MSC handles peak power transients (locomotion bursts, transmission events, destruct initiation) while the nanobattery provides sustained baseline power for control electronics and passive sensing.

5.3 Power Budget Table

6 Locomotion: Biomimetic Spider-Inspired Crawling

6.1 Spider-Inspired Locomotion Principles

Spider legs operate via a hydraulic drive mechanism in which haemolymph (blood) pressure drives leg extension while antagonistic muscles produce flexion—a system achieving remarkably high torque-to-mass ratios [38, 39]. Bionic spider actuators replicating this principle have demonstrated torques

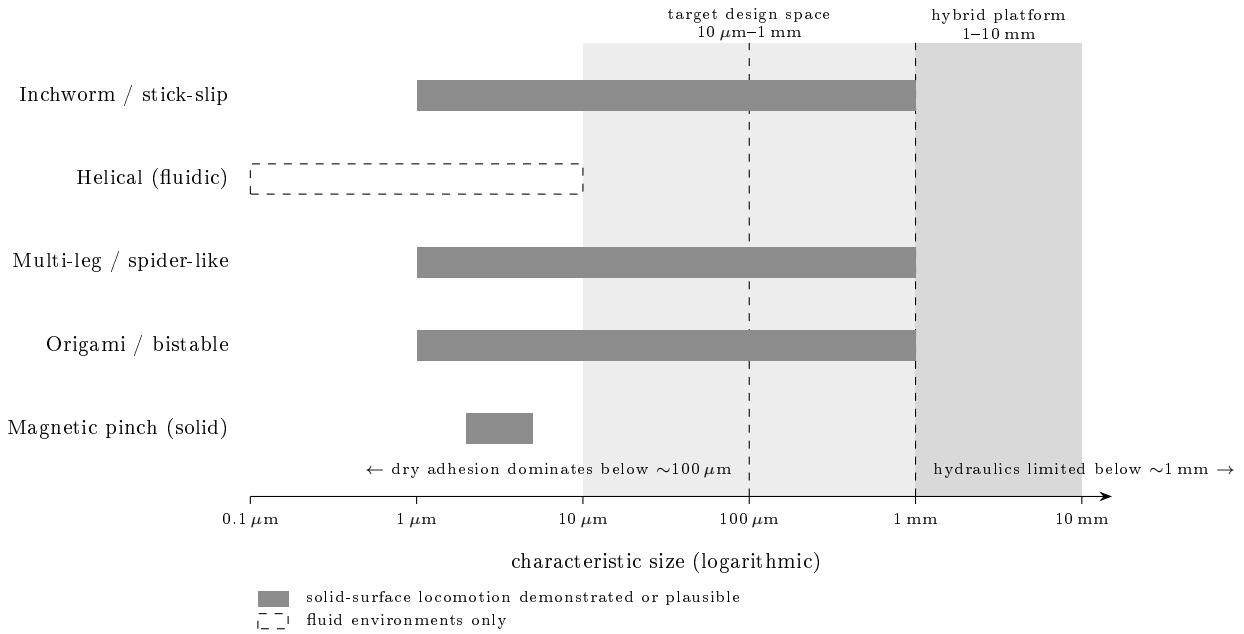


Figure 3. Locomotion mechanism design space: physical scale versus terrain capability, with the target design zone. Bar extents are qualitative, taken from the scale and surface-type entries of Table 2; they are not measured performance ranges.

up to 4.78 N m with stable bidirectional drive [38]. Multi-legged robots (≥ 6 legs) offer inherent high static stability through low centre of mass and wide support polygon, enabling navigation of complex terrain without the balance control overhead required by bipedal or quadrupedal designs [31, 40].

Reinforcement learning-based control frameworks are demonstrating the ability to dynamically adjust body undulation and limb stepping in real time on rugged landscapes, achieving improved speed and adaptability without hand-coded gait programming [40]. This is directly applicable to the nanobot's locomotion controller, which must adapt to the heterogeneous surfaces (soil, rock, concrete, vegetation) encountered in real-world covert deployment.

6.2 Nano-Scale Locomotion Challenges

The target design space for the proposed nanobot crawler (Figure 3) spans approximately 10 μm to 1 mm—a range in which solid-surface locomotion has been demonstrated but presents formidable challenges. Strong dry adhesion dominates at scales below $\sim$100 μm [13], causing micro-scale objects to tenaciously stick to contact surfaces in a way that becomes more pronounced as scale decreases. Brownian motion provides uncontrolled random forcing that disrupts directional locomotion at true nanoscale [35]. Hydraulic actuation—the spider's biological solution—cannot be miniaturised below approximately 1 mm while maintaining adequate force output.

The practical consequence is that the "spider-like" descriptor in the target system is more conceptual (multi-legged stability architecture) than a literal replication of arachnid leg mechanics at nanoscale. The recommended actuation mechanism for the 1–10 mm hybrid platform is a combination of shape memory alloy (NiTi) for primary leg actuation [37], electroelastic elastomers (gelatin/glycerol/CNT/AgNW composites) for compliant joints [36], and magnetic actuators for fine motion control [1].

Table 2. Biomimetic locomotion mechanisms: comparative analysis for nanorobotic surface crawling applications.

Mechanism	Bio-inspiration	Actuation	Demonstrated scale	Surface type	Key nano-scale challenges
Inchworm / stick-slip	Earthworms, caterpillars	Magnetic fields, SMA, electroelastic	Micro to mm	Solid, rough (2–5 μm robots demonstrated)	Strong dry adhesion at nano-scale; precise friction control; continuous power for sustained movement
Helical (fluidic)	Bacterial flagella	Magnetic fields (rotating)	Nano to micro	Fluid environments only	Overcoming Brownian motion; inapplicable to solid-surface crawling; requires external field source
Multi-leg / spider-like	Insects, arachnids	Hydraulic / pneumatic, electroelastic, magnetic	Micro to mm	Solid, uneven, granular	Force generation at scale; control complexity; power for dynamic gaits; miniaturisation of mechanisms
Origami / bistable	Folding patterns in nature	Magnetic fields, voltage-driven SMA	Micro to mm	Confined environments, solid	Robustness under repeated deformation; shape-morphing control complexity; structural fatigue
Magnetic pinch (solid)	Cilia, flagella analogy	Single-axis magnetic field	Nano (2–5 μm)	Solid surfaces (limited)	Very slow; limited force; highly sensitive to surface properties; external magnet dependency

7 Stealth Architecture: Adaptive Camouflage

7.1 Multi-Spectral Concealment Requirements

Effective covert operation in the 21st-century threat environment requires concealment across at least four spectral bands: visible (380–700 nm), near-infrared (700–1400 nm), thermal IR (8–14 μm), and microwave/radar (mm-wave). Modern multi-sensor surveillance systems routinely fuse visual, thermal, and radar data—a "rock" disguise that conceals only visually will be trivially exposed by a thermal camera or ground-penetrating radar. The stealth index $S(t)$ in Eq. (3) must therefore aggregate concealment performance across all relevant spectra.

7.2 Adaptive Camouflage Materials

Graphene-based adaptive surfaces offer the most comprehensive multi-spectral camouflage capability currently demonstrated [43]. By controlling the electron density on graphene's surface, its reflectivity, absorption, and thermal radiation properties are altered across visible, IR, and microwave bands simultaneously. Prototype graphene-pixel wearable jackets have demonstrated effective IR camera concealment by matching background emissivity in real time [43]. Cephalopod-inspired adaptive systems achieve instantaneous cloaking in both visible and IR through pixelised thermal operation [41]. Chameleons demonstrate colour change through tunable guanine nanocrystal lattices [42]; reflecting proteins in cephalopods enable complex pattern formation through self-assembled "reflecting bricks" [42].

For the rock disguise specifically, the nanomaterial engineering challenge is the realistic replication of rock's heterogeneous surface texture at multiple scales—from macroscopic fracture patterns (cm scale) down to mineral grain boundaries (μm scale) and crystal surface roughness (nm scale). Digital image correlation studies demonstrate that homogeneous-textured rocks (marble) are easily distinguished from heterogeneous ones, implying that convincing disguise requires the inherent variability of natural rock—a surface parameterisation challenge best approached through programmable nanomaterial surface texturing [44].

The adaptive aerogel composite framework integrating MXene/reduced graphene oxide with phase-change material and thermochromic coating achieves synchronous visible and IR dual camouflage [30]. Critically, these composites spontaneously increase surface temperature through solar-thermal energy conversion, allowing the object to match high-temperature backgrounds (desert, sun-warmed rock) while simultaneously changing colour—a partial resolution to the stealth–power fundamental conflict of Section 5.

8 Multi-Modal Sensing and Data Architecture

8.1 Sensor Array Specification

The multi-modal sensor array integrates four primary sensing domains as shown in Figure 4. Each domain is implemented at a different physical scale—nanoscale chemical sensors, micro-scale acoustic MEMS, and micro-to-mm optical systems—requiring a hierarchical data fusion architecture that bridges these scales.

Optical. Mantis shrimp-inspired stacked photodiode arrays with metallic nanowire polarisation filters provide simultaneous colour and polarisation sensing [47]. SPRM and iSCAT label-free imaging enable nanoscale object detection without sample modification [50]. Quantum imaging techniques provide sub-shot-noise resolution for covert surveillance at extended ranges [48].

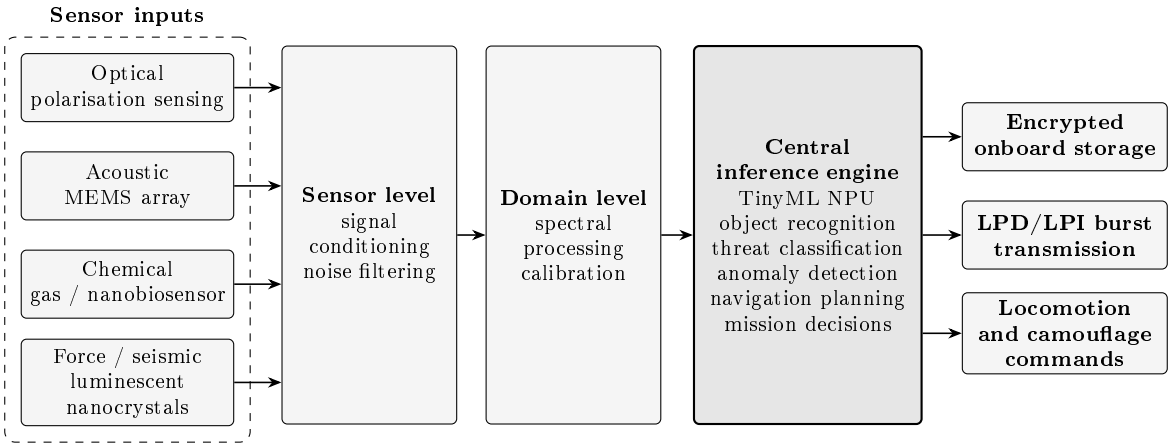


Figure 4. Multi-modal sensor fusion architecture: sensor inputs, hierarchical conditioning and calibration, edge-AI inference, and operational outputs. Inference runs on board; transmission is limited to burst mode.

Acoustic. MEMS microphone arrays with piezoelectric nanogenerator co-integration provide both acoustic sensing and mechanical energy harvesting [29]. Phonon laser (sasar) technology enables highly directional nanoscale acoustic interrogation for structural analysis and covert communication [52]. Acoustic tactile sensors on leg contact surfaces provide terrain classification from wheel/leg deformation patterns [51].

Chemical. Nanostructured metal-oxide gas sensor arrays detect NO_2, SO_2, and PFAS-family compounds at trace concentrations [53, 54]. AI-optimised sensor design is improving specificity and reducing false-positive rates for complex mixed-gas environments [54]. Nanobiosensors with biological recognition elements provide organism-specific detection capability [46].

Force/seismic. All-optical nanoscale force sensors fabricated from luminescent nanocrystals measure mechanical forces remotely by intensity and colour change [49], providing ground vibration data for footstep detection and structural monitoring without electrical contact.

8.2 Edge AI Inference and LPD/LPI Communications

The sensor fusion architecture (Figure 4) implements a hierarchical processing pipeline: raw signal conditioning and noise filtering at the sensor level; spectral processing and calibration at the domain level; and AI-driven classification, anomaly detection, and decision-making at the central inference engine. The edge AI inference engine—implemented on a TinyML NPU with a peak power envelope below 500 mW, supplied by the hybrid storage stage during bursts—performs object recognition, threat classification, navigation planning, and mission decision execution without external communication. Its sustained average draw is bounded by harvested power: at the power densities of Table 1, a $1\,\mathrm{cm}^2$ PV face supplies at most about 5 mW (1–5 $\mathrm{mW/cm}^2$), so inference must be duty-cycled and burst-mode, consistent with the load-shedding logic of Section 4.1.

Communication is the primary stealth vulnerability of the system: any EM transmission can be detected via spectrum scanning [55]. The communication architecture therefore implements a strict LPD/LPI protocol: data is stored onboard in AES-256-encrypted format; transmission occurs only in compressed burst mode (minimising time-on-air); frequency-hopping spread-spectrum minimises spectral detectability; and quantum communication channels are targeted for medium-

Table 3. Multi-modal sensor capabilities: detection principles, scales, applications, and integration challenges.

Sensor type	Detection principle	Scale	Surveillance application	Sensitivity / resolution	Integration challenges
Optical (camera)	Visible light, IR, polarisation	Micro to mm	Visual recon; covert comms decoding	High sensitivity; mantis-shrimp polarisation	Power demand; data processing; covert transmission risk; miniaturisation of optics
Optical (force / imaging)	Force-induced luminescence; light scattering (SPRM, iSCAT)	Nano to micro	Nanoscale force monitoring; label-free single-entity imaging	Very high spatial/temporal resolution	Power; size constraints; data processing bandwidth; integration with locomotion
Acoustic (microphone)	Sound waves; MEMS transduction	Micro to mm	Covert audio surveillance; terrain acoustics	High sensitivity; directional capability	Noise reduction in crawling vibration; EM emanation from wireless TX
Acoustic (phonon / sasar)	Stimulated phonon emission; sound laser	Nano to micro	Structural weakening; covert acoustic intelligence	Unparalleled directionality and precision	Extremely early-stage; power; integration; weaponisation ethics
Chemical (gas / pollutant)	Chemical reactions to electrical signal (metal-oxide, nanobiosensor)	Nano	Environmental monitoring; chemical/biological threat detection	Trace-level detection (ppb); AI-optimised specificity	Sensor drift; cross-sensitivity; stability in field conditions; regeneration
Force / seismic	Luminescent nanocrystals; mechanical deformation	Nano to micro	Ground vibration sensing; footstep detection	Exceptional force sensitivity; remote optical readout	Environmental calibration; temperature sensitivity; substrate coupling

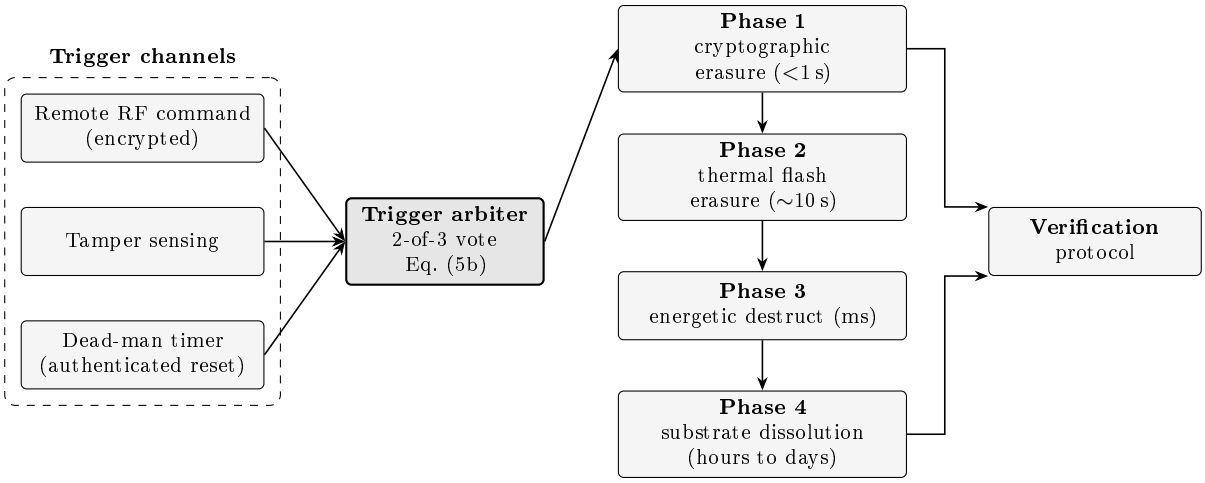


Figure 5. Multi-layer self-destruct cascade: three independent trigger channels feed a 2-of-3 arbiter, which initiates a four-phase sequence followed by a verification protocol. Cascade reliability is modelled by Eq. (5a).

term deployment [48]. The fallback protocol for highest-security scenarios is physical data retrieval—the nanobot caches all collected data and is physically recovered at mission end, with no electronic transmission ever occurring during deployment.

9 Operational Security: Self-Destruct Architecture

9.1 Multi-Phase Destruction Cascade

The self-destruct system implements a four-phase cascade (Figure 5) designed to ensure complete data and physical destruction under the reliability model of Eq. (5a). Phase 1 (cryptographic erasure, $<1\,\mathrm{s}$): the AES-256 private key is irreversibly destroyed, rendering all stored data cryptographically inaccessible. Phase 2 (thermal flash erasure, $\sim 10\,\mathrm{s}$): deliberate heat treatment destroys the charge-trap layer of all onboard flash memory, providing a second, physically separate data destruction mechanism [60]. Phase 3 (energetic destruct, milliseconds): MEMS-scale energetic devices generate GPa-level detonation waves across the physical computing substrate, causing irreversible mechanical destruction of all silicon components [56]. Phase 4 (substrate dissolution, hours to days): cyclododecane sublimation and biodegradable polymer dissolution progressively eliminate the remaining physical evidence of the platform [57].

9.2 Anti-Tampering and Trigger Architecture

The 2-of-3 Byzantine-fault-tolerant trigger arbiter (Eq. 5b) guards against three failure modes: false-positive trigger (unintended self-destruction); false-negative trigger (inability to self-destruct when required); and adversarial suppression (an attacker blocking the destruct command). Three independent trigger channels—encrypted remote RF command, physical tamper detection, and a dead-man timer requiring periodic authenticated reset—are voted upon by the arbiter. Any two channels agreeing initiates the cascade. Anti-tampering features are integrated entirely within the material structure of the platform, avoiding external features that could compromise the rock disguise. Tamper detection uses ambient light sensors detecting case penetration; Hall-effect sensors detecting magnetic signature changes from attempted disassembly; pressure sensors detecting grip forces inconsistent with natural landing; and changes in internal electrical properties of the camou-

Table 4. Secure data destruction methods: mechanisms, applicability, effectiveness, and nanobot deployment challenges.

Method	Mechanism	Applicability	Effectiveness	Speed	Challenges for nanobot deployment
Energetic detonation	GPa-level detonation wave destroys storage chips and circuitry	Flash memory, PCBs, all components	High (irreversible)	Milliseconds	High risk of collateral damage; remote trigger reliability; verification of complete destruction at nano-scale
Thermal erasing	Heat-induced destruction of charge-trap layer in flash memory	Nanoscale flash memory	High	Seconds	Precise heat application; power required for heating element; potential thermal signature exposure
Cryptographic erasure	Encrypt all data; discard private key permanently	Any encrypted memory	High (data only)	<1 second	Requires pre-encryption; key management; physical device remains intact, so must combine with a physical method
Dissolution / sublimation	Biodegradation, chemical dissolution, or cyclododecane sublimation of substrate	Transient electronics on compatible substrates	High (structural)	Minutes to ∼12 days	Environmental dependency; reaction rate control; material compatibility with other components
Data overwriting	Random pattern written multiple times over existing data	Addressable storage (limited SSD efficacy)	Moderate–high	Minutes to hours	Time-consuming; SSD wear-leveling may leave inaccessible remnants; not suitable as primary method
Cooling-triggered dissolution	Conductors dissolve when cooled below critical temperature	Specific conductor materials	Moderate	Seconds to minutes	Temperature control mechanism; not universally applicable; environmental triggering risk

flage layer upon mechanical deformation [62, 63]. All tamper sensors feed directly to the trigger arbiter.

10 Prototype Hardware: Image Placeholder Section

The following figure spaces are reserved for hardware prototype images to be inserted as device development progresses. These sections document the physical realisation of the nanobot surveillance crawler system from initial laboratory prototype through to field-capable units.
Note. Blueprint-level design drawings and detailed schematic documentation are retained as proprietary intellectual property and are not included in this publication. The prototype image placehold-

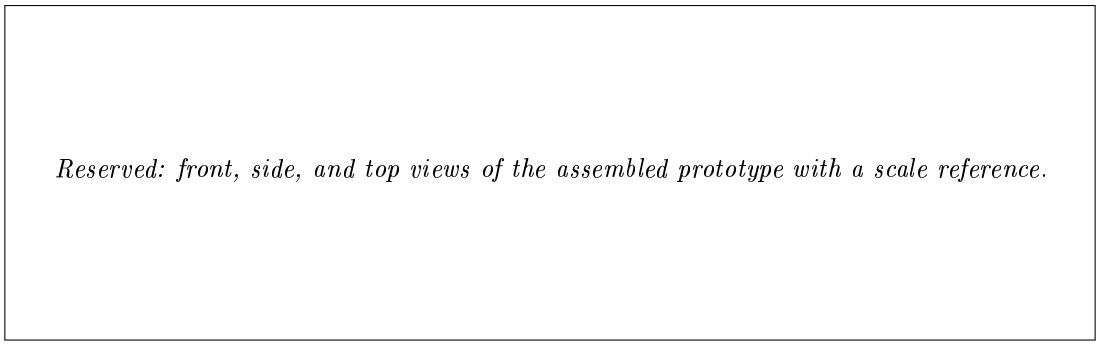


Figure P.1. External assembly: prototype v1 (camouflage shell, leg positions, sensor apertures where visible).

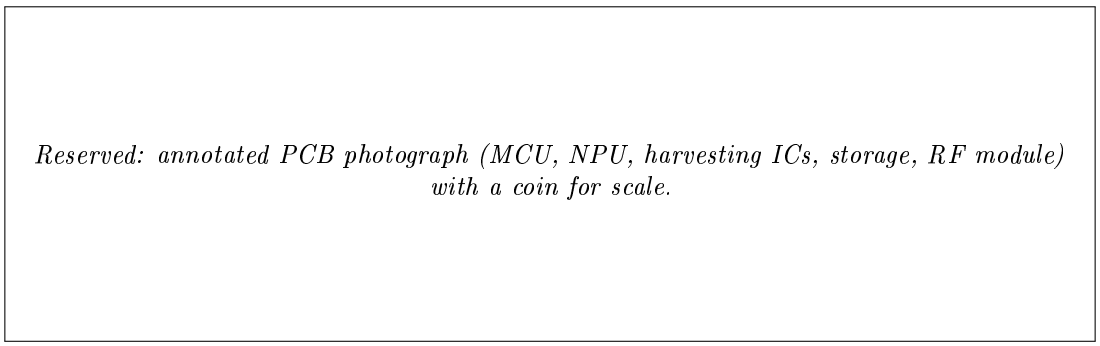


Figure P.2. Internal PCB and power module.

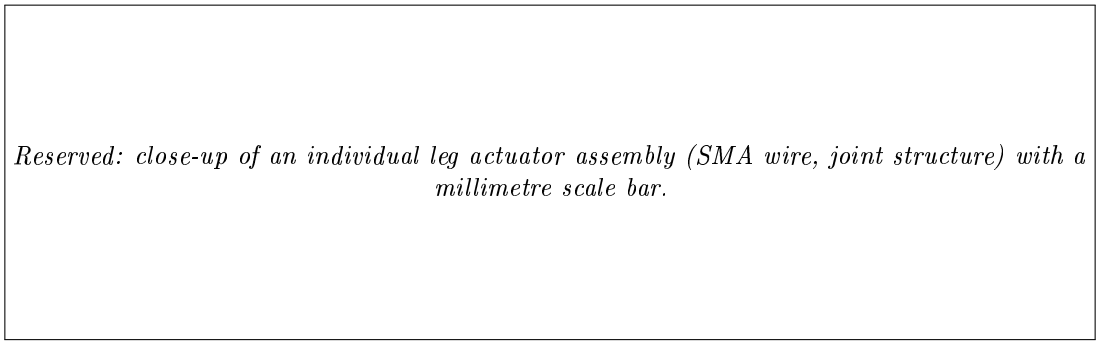


Figure P.3. Leg actuator assembly detail (SMA / electroelastic mechanism).

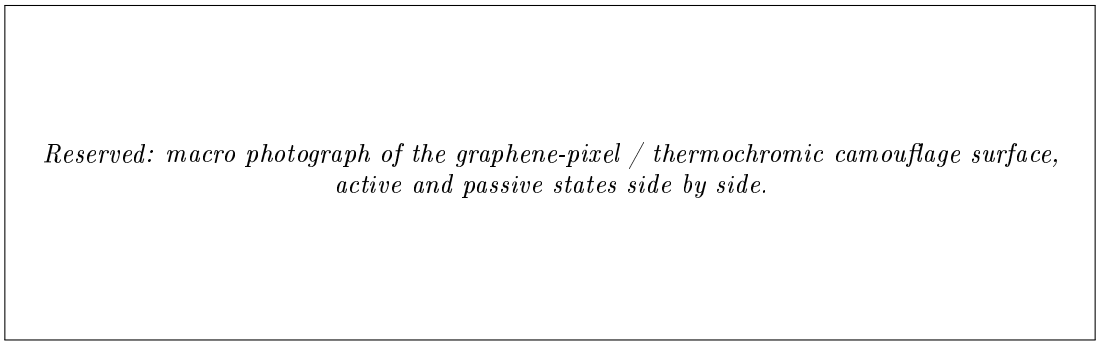


Figure P.4. Adaptive camouflage surface detail: active and passive (rock-mimicking) states.

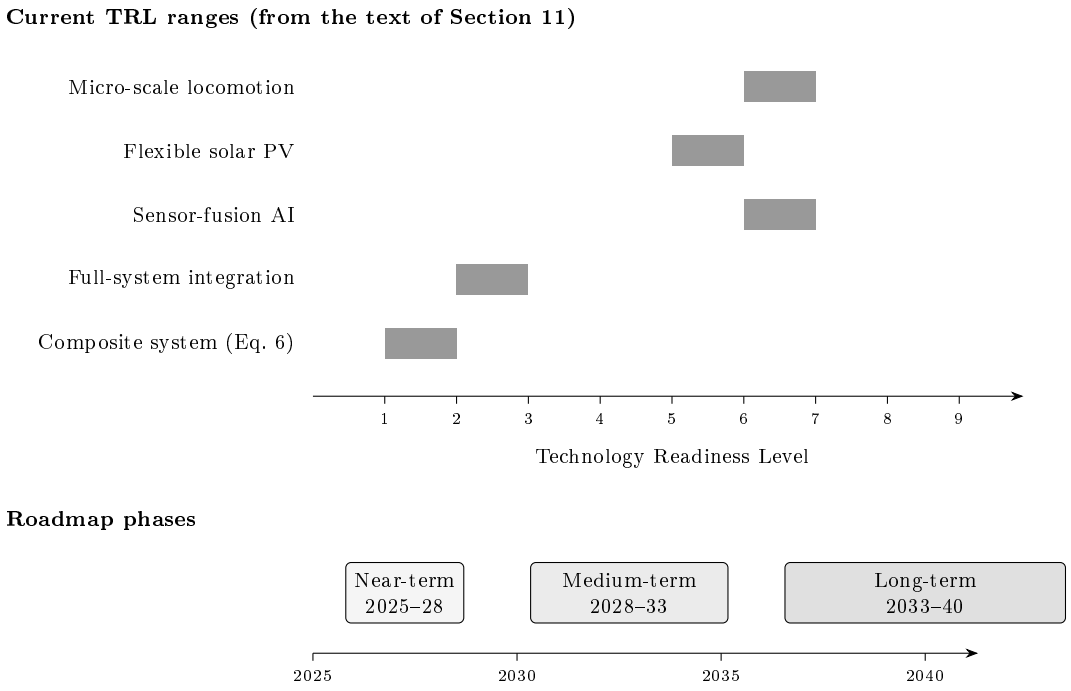


Figure 6. Technology readiness: current subsystem TRL ranges (component tracks versus full-system integration) and the three roadmap phases of Sections 11.2–11.4. The diagram is schematic; bars show only the ranges stated in the text, not a fitted trajectory.

ers above are intended to document physical hardware evidence of system development for research and funding purposes only.

11 Integration Challenges and Technology Roadmap

11.1 The Fundamental Integration Bottleneck

The Technology Readiness Level analysis (Figure 6) reveals the defining characteristic of this design challenge: individual component technology tracks—micro-scale locomotion (TRL 6–7), flexible solar PV (TRL 5–6), sensor-fusion AI (TRL 6–7)—are all advancing rapidly toward deployment readiness. However, full-system integration (TRL 2–3) lags all component tracks by 2–4 TRL levels, and this gap is not merely temporal but reflects fundamental multi-scale integration challenges that are not automatically resolved by individual component advancement.

The integration penalty $\Delta_{\text{integration}}$ in Eq. (6) captures this deficit. The primary contributors are: interface incompatibilities between components operating at different physical scales (nm sensors $\rightarrow$ μm actuators $\rightarrow$ mm power $\rightarrow$ cm communications); power delivery architectures that must serve components with fundamentally different voltage and current profiles; thermal management across subsystems with opposing thermal requirements (PV harvesting prefers heat; actuation electronics prefer cool operation; TEG harvesting requires a maintained gradient); and the emergent behaviours of integrated systems containing large numbers of nanocomponents that are not predictable from individual component characterisation [16].

Table 5. Critical integration challenges: technical detail, severity, and mitigation strategy.

Challenge	Technical detail	Severity	Mitigation strategy
Power autonomy at nano-scale	Sufficient continuous power for locomotion + sensing + processing + comms + self-destruct simultaneously. Miniature PV output is fundamentally limited; MSC energy density insufficient for sustained active missions.	Critical	Hybrid multi-modal harvesting (PV + piezo + TEG); adaptive load shedding; burst-mode operation; nano-battery hybrid storage; ultra-low-power component design ($<1\,\mu$W idle)
Robust solid-surface locomotion	Spider-like multi-legged crawling on rough, irregular terrain at true nano-scale. Strong dry adhesion worsens at smaller scales; hydraulic spider mechanisms cannot be miniaturised to nanoscale.	Critical	Hybrid micro-nano architecture ($\sim$1 mm functional scale); SMA + electroelastic actuators; RL-based gait control; vdW/electrostatic adhesion modulation
Adaptive multi-spectral camouflage	Dynamic visual + thermal + microwave concealment while integrating PV power sources. The efficiency vs. covertness trade-off is fundamental: better disguise means less solar power.	High	Graphene-based adaptive pixels (visible + IR); thermochromic + MXene aerogel composites; segmented solar-thermal surfaces; active emissivity matching
Covert multi-sensor integration	Miniaturising optical, acoustic, chemical, and force sensor arrays with onboard AI fusion processing without detectable EM emanations from operation or transmission.	High	Hierarchical sensing (wide-field low-res + narrow-field high-res); LPD/LPI burst RF TX; quantum communication channels; on-board TinyML processing; store-carry-retrieve data protocol
Reliable on-demand self-destruction	Complete, irreversible destruction of all data and physical components on demand without unintended activation, collateral damage, or detectable pre-destruct signature.	High	Multi-phase cascade (crypto erase $\rightarrow$ thermal erase $\rightarrow$ energetic destruct $\rightarrow$ substrate dissolution); BFT trigger arbiter; 2-of-3 trigger consensus; remote verification protocol
Multi-scale system integration	Integrating nanoscale sensors, micro-scale actuators, and mm-scale power/comms components into a single cohesive unit with compatible interfaces, power delivery, and data pathways.	Critical	Digital thread design methodology; hybrid micro-nano architecture as practical stepping stone; modular subsystem interfaces; co-design of power and data pathways across scales
Covert data transmission	Wireless transmission is detectable via spectrum scanning. Any EM emanation, even passive, can compromise position.	High	LPD/LPI spread-spectrum; quantum entanglement channels (research); acoustic covert channel; time-delayed burst TX; physical data retrieval as fallback
Regulatory and ethical governance	No existing framework governs autonomous, covert, self-destructing nanoscale surveillance agents. Dual-use proliferation risk is severe once core technologies mature.	Critical	Proactive international governance frameworks; ICRC / UNICRI engagement; export controls on enabling technologies; mandatory ethics review for dual-use nano-ISR research

11.2 Near-Term Roadmap (2025–2028)

Roadmap years are counted from the May 2025 literature cut-off of this review (Section 3).

1. Demonstrate a sub-10 mm hybrid micro-nano crawling platform with SMA + electroelastic actuation achieving >10 body-lengths/second on rough terrain with reinforcement-learning gait control.
2. Integrate flexible perovskite PV (>20% efficiency) with a thermochromic camouflage layer in a single adaptive surface tile; characterise the efficiency vs. concealment Pareto front.
3. Demonstrate the four-phase self-destruct cascade on a representative electronics assembly; measure cascade reliability R_k per phase and validate the Eq. (5a) model.
4. Implement TinyML sensor fusion (optical + acoustic + chemical) on an embedded platform with a sub-100 mW peak draw; benchmark classification accuracy against a full-scale deep-learning baseline.

11.3 Medium-Term Roadmap (2028–2033)

1. Full subsystem integration at 5–10 mm platform scale; characterise composite operational effectiveness $E(t)$ (Eq. 3) under representative operational scenarios.
2. Demonstrate LPD/LPI burst communication achieving $< 10^{-6}$ probability of detection per transmission event via spread-spectrum + temporal randomisation.
3. Validate the camouflage adaptation control law (Eq. 4) in real-time operation against a multi-spectral sensor array (visible + IR + radar).
4. Field trial in representative uncontrolled terrain: demonstrate sustained autonomous operation for >4 hours at target TRL 5 performance level.

11.4 Long-Term Vision (2033–2040)

1. True nano-scale functional prototype (100–500 μm): demonstrate solid-surface locomotion, onboard sensing, and wireless data transmission at the sub-millimetre scale.
2. Batch manufacture capability: demonstrate consistent production of >100 identical units from the digital design thread with <5% performance variance.
3. Swarm coordination: demonstrate autonomous multi-unit collaborative sensing with distributed task allocation, using the AMSCA coordination framework [69] (companion paper).
4. Regulatory demonstration: operate a deployed unit under an agreed governance framework with real-time oversight capability, supporting development of international regulatory standards.

12 Discussion

12.1 The Scale–Efficiency Paradox as Fundamental Constraint

The most important finding of this analysis is that the challenges facing the covert nanorobotic crawler are not merely engineering difficulties awaiting incremental solution—several represent fundamental physical trade-offs that cannot be resolved simultaneously by any single design. The

stealth–power conflict is the clearest example: a surface optimised for covert visual blending into a rock background is geometrically and optically incompatible with a surface optimised for solar energy harvesting. These two objectives are Pareto-non-dominated in the design space—improvement in one can only come at the cost of the other.

Similarly, the autonomy–covertness trade-off at the information level—high autonomous decision-making requires substantial computation, which generates detectable heat and EM emissions, compromising the very stealth the autonomy is designed to protect—has no clean resolution within current technology. The LPD/LPI communication architecture and store-carry-retrieve data protocol represent partial mitigations, not solutions. The formal treatment of these trade-offs in Eqs. (1)–(3) makes explicit what qualitative assessments only imply: there exist hard Pareto boundaries in the design space, and successful system design requires explicit management of trade-offs rather than optimising each subsystem independently.

12.2 The Hybrid Micro-Nano Architecture as Practical Path

Given the fundamental barriers to true nanoscale integration, the hybrid micro-nano architecture—functional scale 1–10 mm with nanoscale internal components—represents the most credible near-term development path. This is not a compromise but a recognition that the relevant capabilities (covert locomotion, multi-modal sensing, adaptive camouflage, self-destruct) are achievable at the micro-scale with nanoscale-enhanced performance, while true nanoscale integration awaits foundational advances in multi-scale assembly, power delivery, and embedded computation. The digital design thread methodology, developed for the AMSCA swarm framework [69] (companion paper), applies directly to the nanobot manufacturing workflow: parametric design models linked through simulation, fabrication, and test stages enable rapid iteration and quality assurance across the challenging tolerances of micro-scale fabrication.

12.3 Limitations of This Analysis

This paper presents a conceptual framework and mathematical model; no hardware prototype has been built, and the models have not been experimentally validated. The power budget values in Eq. (1) are parameterised from literature values for individual components and may not accurately reflect integrated system performance due to parasitic losses and interface inefficiencies. The TRL integration penalty estimate ($\Delta_{\text{integration}} \approx 2$) is based on analogous system classes and may differ for this specific combination of subsystems. Mathematical models are proposed as analytical scaffolding for future experimental design, not empirically fitted models.

13 Ethical, Legal, and Governance Analysis

13.1 Privacy and Human Rights Implications

A fully functional covert nanorobotic surveillance system would represent an unprecedented threat to privacy. Unlike existing surveillance technologies—CCTV cameras, drones, planted listening devices—a genuinely undetectable and self-destructing nanobot leaves no physical evidence of its presence or operation. The ability of nanomachines to perform undetectable surveillance through molecular-sized sensors could fundamentally jeopardise the right to privacy as protected under Article 12 of the Universal Declaration of Human Rights and analogous national frameworks [6]. The proliferation of concealed IoT devices already poses significant threats [55]; a nanorobotic

capability operating at orders of magnitude smaller scale and with active disguise capability would make even the most sophisticated counter-surveillance sweeps ineffective.

13.2 Accountability Gap in Autonomous Covert Systems

The intersection of autonomy (Level 4–5, no real-time human direction) and covertness (untraceable, self-erasing) creates a profound accountability vacuum. If a covert nanobot malfunctions, causes unintended harm, or is used for unlawful surveillance, attribution becomes effectively impossible when the device has self-destructed. The existing legal frameworks for accountability in autonomous military and intelligence systems [64, 65] assume that a physical record of action can be examined. Self-destruction eliminates this record. Any governance framework for covert nanorobotic systems must address this gap pre-emptively rather than reactively.

13.3 Dual-Use Proliferation Risk

Many of the enabling technologies for the proposed system—flexible PV, MEMS sensors, adaptive materials, TinyML inference, miniature RF modules—are commercially available (miniature and flexible PV cells [18, 19]; TinyML inference on commodity microcontrollers [14]) or developed in dual-use research programmes [66]. As individual component TRLs advance and costs fall, the barrier to assembling a capable covert surveillance platform will decrease. Once core technologies mature, they could proliferate to non-state actors, criminal organisations, and authoritarian regimes for domestic surveillance applications [67]. The proliferation risk is compounded by the difficulty of export controlling technologies whose primary commercial applications (IoT, wearables, medical devices) are entirely civilian.

13.4 Recommended Governance Framework

International classification framework. Establish an agreed international categorisation of nanorobotic surveillance capabilities by function and scale, analogous to the Wassenaar Arrangement for conventional dual-use technologies, with export controls on key enabling technologies above defined capability thresholds.

Mandatory ethics review. Require independent ethics review for any research programme combining autonomous locomotion, covert camouflage, surveillance sensing, and self-destruct capability, regardless of stated application.

Red-line prohibitions. Impose an international prohibition on deployment of self-destructing surveillance nanobots in civilian environments without prior judicial authorisation, analogous to rules governing covert entry and electronic surveillance.

Transparency requirements. Subject any state programme developing covert nanorobotic surveillance capability to parliamentary/legislative oversight with classified but auditable records, preventing the total accountability vacuum that self-destruct capability otherwise enables.

ICRC/UNICRI engagement. Engage proactively with the International Committee of the Red Cross and the UN Interregional Crime and Justice Research Institute to develop specific guidance on nanorobotic systems before deployment capability is achieved.

14 Conclusions

This paper has presented a technical, mathematical, and ethical analysis of the covert nanorobotic surveillance crawler concept. The central finding is unambiguous: a fully functional system integrating solar-powered autonomous locomotion, adaptive multi-spectral camouflage, multi-modal sensing, covert communications, and on-demand self-destruct at the true nanoscale is not achievable with current technology, and faces fundamental physical trade-offs—particularly the stealth–power conflict and the locomotion miniaturisation wall—that will constrain design choices even as individual component technologies advance.

The formal mathematical framework introduced here—power budget optimisation (Eqs. 1–2), composite operational effectiveness (Eq. 3), camouflage adaptation control law (Eq. 4), self-destruct cascade reliability (Eq. 5), and TRL integration convergence bound (Eq. 6)—provides quantitative tools for research investment decisions, subsystem requirement derivation, and performance prediction that qualitative feasibility assessments cannot offer. The hybrid micro-nano architecture, targeting a 1–10 mm functional scale with nanoscale internal components, is identified as the practical engineering path: achieving the required capability profile within current and near-term technology constraints while providing a clear progression toward genuine nanoscale realisation.

The six architectural diagrams, five data tables, and prototype image placeholder framework provide a complete specification structure for programme development, with the prototype sections reserved for hardware images as development proceeds. The roadmap targets TRL 5 (validation in a relevant environment) by approximately 2031 and TRL 7 (system prototype in operational environment) by approximately 2037, conditional on focused investment in the identified bottleneck areas: multi-scale integration methodology, power budget management, and LPD/LPI communication protocols.

The ethical and governance analysis establishes that this technology must not be developed without commensurate development of international governance frameworks. The combination of autonomous operation, covert disguise, and self-erasing capability creates an accountability vacuum with no precedent in existing surveillance law. Proactive governance—classification frameworks, mandatory ethics review, red-line prohibitions, parliamentary oversight—must be developed before deployment capability is achieved, not after. The scientific community developing the enabling technologies carries a particular responsibility to lead this governance conversation rather than leaving it to emerge reactively from deployment controversies.

Conflict of Interest

The author is affiliated with Drooid. No other competing interests are declared. This paper presents a feasibility analysis and conceptual framework for research purposes. No operational surveillance system is described, proposed, or endorsed.

Acknowledgements

The author acknowledges the research communities across nanorobotics, materials science, MEMS, adaptive optics, and autonomous systems whose published work this synthesis draws upon.

Data Availability

This paper presents a conceptual framework and literature synthesis. No primary experimental datasets were generated. All referenced performance data are sourced from the cited primary literature. Mathematical models are original derivations available for reproduction from the equations presented.

References

[1] Nanorobotics. Wikipedia, accessed May 2025. https://en.wikipedia.org/wiki/Nanorobotics

[2] Advancements in micro/nanorobots in medicine. *ACS Omega*, 2024. https://pubs.acs.org/doi/10.1021/acsomega.4c09806

[3] K. Lund, A. J. Manzo, N. Dabby, et al. Molecular robots guided by prescriptive landscapes. *Nature*, 465(7295):206–210, 2010. https://doi.org/10.1038/nature09012

[4] Nanorobotics: Theory, applications, how does it work? Built In, 2025.

[5] Small security: Nanotechnology and future defense. DTIC Report ADA422476, 2004.

[6] The ethics of nanotechnology. Markkula Center for Applied Ethics, Santa Clara University.

[7] Micro-/nanoscale robotics for chemical and biological sensing. PMC10530003, 2023.

[8] K. Y. Ma, P. Chirarattananon, S. B. Fuller, and R. J. Wood. Controlled flight of a biologically inspired, insect-scale robot. *Science*, 340(6132):603–607, 2013. https://doi.org/10.1126/science.1231806

[9] Untethered swarm robots with independent crawling and rolling motions. CORE, 2023.

[10] Untethered bistable origami crawler for confined applications. ResearchGate, 2024.

[11] Researchers have created the world's smallest walking robot. Tomorrow's World Today, 2024.

[12] Smallest walking robot makes microscale measurements. Cornell Chronicle, 2024.

[13] Nanoscale Lamb wave-driven motors in nonliquid environments. PMC6408156, 2019.

[14] P. P. Ray. A review on TinyML: State-of-the-art and prospects. *J. King Saud Univ. Comput. Inf. Sci.*, 34(4):1595–1623, 2022. https://doi.org/10.1016/j.jksuci.2021.11.019

[15] Energy harvesting in nanosystems: Powering the next generation of IoT. *Frontiers in Nanotechnology*, 2021.

[16] Research challenges for integrated systems nanomanufacturing. Northeastern University, 2023.

[17] Internet of Nano-Things (IoNT): A comprehensive review. PMC10007150, 2023.

[18] Miniature solar panel cell breakout board. Questwise Ventures / Tindie, 2024.

[19] Flexible solar panel (1.5 V 250 mA). DFRobot product documentation, 2024.

[20] The truth about flexible solar panels: Pros, cons & when they make sense. Moser Baer Solar, 2024.

[21] New nanoscale method could improve solar cell efficiency. Croucher Foundation, 2024.

[22] What are covert solar panels? Nexgent News, 2024.

[23] Q. Tai and F. Yan. Emerging semitransparent solar cells: Materials and device design. *Adv. Mater.*, 29(34):1700192, 2017. https://doi.org/10.1002/adma.201700192

[24] Maximizing underwater energy harvesting efficiency using flexible solar cells. *PNAS*, 2025. https://doi.org/10.1073/pnas.2423651122

[25] On-chip micro-supercapacitor with high areal energy density. PubMed 38497508, 2024.

[26] Nanotechnology-based batteries: A new frontier for enhanced energy storage. IDSTCH, 2024.

[27] Piezoelectric-based microrobots for locomotion. ResearchGate, 2023.

[28] Thermoelectric generator. Wikipedia, accessed May 2025.

[29] Piezoelectric nanogenerators derived self-powered sensors. BINN/CAS, 2021.

[30] Adaptive MXene/reduced graphene oxide hybrid aerogel composites for dual camouflage. *ACS Nano*, 2023. https://doi.org/10.1021/acsnano.3c00573

[31] What is a spider robot and how does it work? Standard Bots, 2025.

[32] Biomimetics. Wikipedia, accessed May 2025.

[33] J. F. V. Vincent, O. A. Bogatyreva, N. R. Bogatyrev, A. Bowyer, and A.-K. Pahl. Biomimetics: its practice and theory. *J. R. Soc. Interface*, 3(9):471–482, 2006. https://doi.org/10.1098/rsif.2006.0127

[34] Biomimetics micro robot with neural network locomotion. ResearchGate, 2014.

[35] Autonomous nanorobots as miniaturized surgeons. PMC11013175, 2024.

[36] Electromagnetic-driven spider-inspired soft robot. ResearchGate, 2024.

[37] Actuation mechanisms and applications for soft robots: A review. *MDPI Applied Sciences*, 2023.

[38] A spider-joint-like bionic actuator with triangular prism shape. *MDPI Biomimetics*, 2023.

[39] D. A. Parry and R. H. J. Brown. The hydraulic mechanism of the spider leg. *J. Exp. Biol.*, 36(2):423–433, 1959.

[40] Learning to enhance multi-legged robot on rugged landscapes. *arXiv*:2409.09473, 2024.

[41] Research progress of bionic adaptive camouflage materials. *Frontiers in Materials*, 2021.

[42] Research progress of bionic adaptive camouflage materials (PDF). *Frontiers in Materials*, 2021.

[43] The genius stealth inventions making people, planes and cities invisible. BBC Science Focus, 2024.

[44] Effects of rock texture on digital image correlation. *MDPI Geosciences*, 2025.

[45] Nanoscale materials in biomedical applications of sensors: Insights from a comprehensive landscape analysis. *ACS Appl. Nano Mater.*, 8(3):1305–1331, 2025. https://doi.org/10.1021/acsanm.4c04463

[46] Nano biosensors for prediction of environmental pollutants. ResearchGate, 2024.

[47] Mantis shrimp-inspired camera. Illinois News Bureau, 2024.

[48] Making the invisible, visible: Quantum surveillance. QUANTIC, 2024.

[49] New all-optical nanoscale sensors of force. Tech Explorist, 2024.

[50] Label-free optical imaging of nanoscale single entities. PMC10990724, 2024.

[51] Acoustic tactile sensing for mobile robot wheels. *arXiv*:2402.18682, 2024.

[52] Sound lasers: Revolutionizing acoustic technology with military applications. IDSTCH, 2024.

[53] Nanostructured gas sensors: From air quality to healthcare. PMC8398721, 2021.

[54] AI helps scientists design better sensors for pollutants. UChicago PME, 2024.

[55] Securing the invisible: Hidden tech threatens national security. Bitter Winter, 2024.

[56] Research on energetic micro-self-destruction devices. *Micromachines*, 14(5):961, 2023.

[57] Now you see it, now you don't: Self-destructing electronics. Advanced Science News, 2023.

[58] Cooling-triggered self-destructing electronics. Vanderbilt / CTTC, 2024.

[59] What is data sanitization? Imperva, 2024.

[60] Nanoscale flash research articles. R Discovery, 2024.

[61] Fail-safes in nanotechnology. Wikipedia, accessed May 2025.

[62] Design ideas for tamper detection circuits. Planet Analog, 2024.

[63] Sophisticated anti-tamper solutions and countermeasures. Secure-IC, 2024.

[64] Autonomous military robotics: Risk, ethics, and design. Cal Poly / ONR, 2013.

[65] Losing humanity: The case against killer robots. Human Rights Watch, 2012.

[66] Army completes autonomous micro-robotics research program. US Army, 2018.

[67] Security implications of synthetic biology and nanobiotechnology. UNICRI, 2012.

[68] National Aeronautics and Space Administration, Earth Science Technology Office. Technology readiness levels (TRLs). `https://esto.nasa.gov/trl`

[69] D. Kalu. AI-native generative design pipeline for autonomous microsystems: A unified computational framework for swarm-native autonomous systems (AMSCA). Manuscript, Drooid. `https://humancyyborg.github.io/port-files/publications/04_swarm_robotics_loop.pdf`